

Measurement rate from input–output theory

Student notes

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1 Model and goal

We want to know how fast a reflected microwave signal tells us whether a system qubit S is in $|0\rangle$ or $|1\rangle$. S is not looked at directly. It is coupled to a detector qubit D, which is coupled to a cavity, and we probe the cavity in reflection through one port.

We set $\hbar = 1$ and work in a frame rotating at the drive frequency ω_d . The Hamiltonian is

$$H = \Delta_r a^\dagger a + \frac{\Delta_D}{2} \sigma_z^D + \frac{\Delta_S}{2} \sigma_z^S + g \left(a^\dagger \sigma_-^D + a \sigma_+^D \right) + J_L \sigma_z^S \sigma_z^D + J_\perp \left(\sigma_+^S \sigma_-^D + \sigma_-^S \sigma_+^D \right) + i \left(\varepsilon a^\dagger - \varepsilon^* a \right). \quad (1)$$

Here $\Delta_r = \omega_r - \omega_d$, and Δ_D, Δ_S are the qubit detunings from ω_d . The last term is the coherent drive; Sec. 2 shows how ε is related to the incoming microwave amplitude.

The cavity loses photons at rate κ , all of them into the port we measure (no internal losses). The qubits are subject to pure dephasing only (no relaxation). The Lindblad equation is

$$\dot{\rho} = -i[H, \rho] + \kappa \mathcal{D}[a]\rho + \frac{\gamma_D}{2} \mathcal{D}[\sigma_z^D]\rho + \frac{\gamma_S}{2} \mathcal{D}[\sigma_z^S]\rho \equiv \mathcal{L}\rho, \quad \mathcal{D}[c]\rho = c\rho c^\dagger - \frac{1}{2}\{c^\dagger c, \rho\}. \quad (2)$$

With this normalization, γ_D and γ_S are the rates at which the qubit coherences decay, $\langle \sigma_- \rangle \propto e^{-\gamma t}$. Dephasing commutes with σ_z , so it never flips a qubit. Qubit relaxation and internal cavity losses can be added later as extra dissipators; nothing below depends on the exact form of $\mathcal{L}$.

The question. Prepare S in $|j\rangle$ ($j = 0$ or 1), with D in its ground state and the cavity empty, $|g\rangle\langle g|_D \otimes |0\rangle\langle 0|$ (a cold detector; other initial states of D are discussed in Sec. 4). Record the reflected field for a time τ . How large must τ be before the two records can be told apart? The answer defines the measurement time τ_m and the measurement rate Γ_m . The goal of these notes is a formula for Γ_m that needs only (i) input–output theory and (ii) the Lindblad equation (2), with no approximation on H .

2 Input–output relations

The port is a transmission line carrying a continuum of modes. Its incoming and outgoing parts are described by field operators $b_{\text{in}}(t)$ and $b_{\text{out}}(t)$, normalized so that $b^\dagger b$ is a photon flux (photons per second). Input–output theory (Gardiner–Collett) gives a Heisenberg–Langevin equation for the cavity field and a boundary condition linking input, output and cavity:

$$\dot{a} = -i[a, H] - \frac{\kappa}{2}a - \sqrt{\kappa}b_{\text{in}}(t), \quad b_{\text{out}}(t) = b_{\text{in}}(t) + \sqrt{\kappa}a(t). \quad (3)$$

Splitting off the drive. The incoming microwave is a coherent tone plus vacuum noise. In the rotating frame we write $b_{\text{in}} \rightarrow \beta_{\text{in}} + b_{\text{in}}$, with β_{in} a c-number and the new b_{in} in the vacuum state. Inserting this into (3), the β_{in} term is exactly what the drive term in (1) produces, with $\varepsilon = -\sqrt{\kappa} \beta_{\text{in}}$. So the coherent drive lives in H , and b_{in} is pure vacuum from now on. The reflected field contains $\beta_{\text{in}} + \sqrt{\kappa} \langle a \rangle$. The β_{in} part is the same for S in $|0\rangle$ and $|1\rangle$, so it drops out of every signal difference below.

Vacuum input. Everything we need about the noise is contained in

$$[b_{\text{in}}(t), b_{\text{in}}^\dagger(t')] = \delta(t - t'), \quad \langle b_{\text{in}}(t) b_{\text{in}}^\dagger(t') \rangle = \delta(t - t'), \quad \langle b_{\text{in}}^\dagger(t) b_{\text{in}}(t') \rangle = \langle b_{\text{in}}(t) b_{\text{in}}(t') \rangle = 0. \quad (4)$$

In words: $b_{\text{in}}|\text{vac}\rangle = 0$ and $\langle \text{vac}|b_{\text{in}}^\dagger = 0$. The only non-zero correlator is the anti-normally ordered one.

Causality. For any system operator A (cavity or qubit) at time t :

1. Future input has not reached the system yet: $[A(t), b_{\text{in}}(t')] = [A(t), b_{\text{in}}^\dagger(t')] = 0$ for $t' > t$.
2. Past output has already left and cannot be changed: $[A(t), b_{\text{out}}(t')] = [A(t), b_{\text{out}}^\dagger(t')] = 0$ for $t' < t$.

Combining 2 with the boundary condition (3) gives the identity used in Sec. 5:

$$[A(t), b_{\text{in}}^\dagger(t')] = -\sqrt{\kappa} [A(t), a^\dagger(t')], \quad t > t'. \quad (5)$$

One more fact. The output field obeys the same commutator as the input, $[b_{\text{out}}(t), b_{\text{out}}^\dagger(t')] = \delta(t - t')$, because the overall evolution is unitary.

3 What we measure

The reflected field is amplified and mixed with a local oscillator of phase φ (homodyne detection). The detector then measures the output quadrature

$$X_{\text{out}}(t) = e^{-i\varphi} b_{\text{out}}(t) + e^{i\varphi} b_{\text{out}}^\dagger(t) = X_{\text{in}}(t) + \sqrt{\kappa} X_\varphi(t), \quad (6)$$

with the vacuum quadrature $X_{\text{in}} = e^{-i\varphi} b_{\text{in}} + e^{i\varphi} b_{\text{in}}^\dagger$ and the cavity quadrature $X_\varphi = e^{-i\varphi} a + e^{i\varphi} a^\dagger$. (We have dropped the known constant from β_{in} .)

The simplest way to decide between $|0\rangle$ and $|1\rangle$ is to integrate the signal up to time τ and compare with a threshold. The integrated signal is the operator

$$M(\tau) = \int_0^\tau X_{\text{out}}(t) dt. \quad (7)$$

Is M a proper observable, given that it is a sum of operators at different times? Yes: output quadratures at different times commute,

$$[X_{\text{out}}(t), X_{\text{out}}(t')] = [b_{\text{out}}(t), b_{\text{out}}^\dagger(t')] - [b_{\text{out}}(t'), b_{\text{out}}^\dagger(t)] = \delta(t - t') - \delta(t' - t) = 0. \quad (8)$$

So all the $X_{\text{out}}(t)$ can be measured jointly, and $M(\tau)$ is their sum. For each initial S state j it has a mean $\langle M \rangle_j$ and a variance σ_j^2 . These two numbers are all we need for a signal-to-noise ratio.

4 The signal

The mean signal needs only the Lindblad equation. Take the expectation of $M(\tau)$ with S initially in $|j\rangle$. The vacuum part averages to zero, $\langle X_{\text{in}}(t) \rangle = 0$, so

$$\langle M(\tau) \rangle_j = \sqrt{\kappa} \int_0^\tau \langle X_\varphi(t) \rangle_j dt, \quad \langle X_\varphi(t) \rangle_j = 2 \text{Re} [e^{-i\varphi} \text{Tr}\{a \rho_j(t)\}], \quad (9)$$

where $\rho_j(t)$ solves (2) from $\rho_j(0) = |j\rangle\langle j|_S \otimes |g\rangle\langle g|_D \otimes |0\rangle\langle 0|_{\text{cav}}$.

Why the Lindblad equation is allowed here. The Heisenberg–Langevin equations (3), averaged over vacuum inputs, give exactly the same one-time averages as the Lindblad equation. This is the standard equivalence: the dissipator $\kappa\mathcal{D}[a]$ is what is left of the bath once its (vacuum) input is traced out.

The quantity that distinguishes the two S states is the difference

$$\Delta X(t) = \langle X_\varphi(t) \rangle_1 - \langle X_\varphi(t) \rangle_0, \quad \Delta M(\tau) = \langle M(\tau) \rangle_1 - \langle M(\tau) \rangle_0 = \sqrt{\kappa} \int_0^\tau \Delta X(t) dt. \quad (10)$$

Two regimes are worth keeping in mind:

- **QND-like.** If S does not flip, $\rho_j(t)$ relaxes to a steady state that still remembers j . Then $\Delta X(t)$ tends to a constant after a ring-up time of order $1/\kappa$, and ΔM grows linearly in τ .
- **Non-QND.** If S can flip (via $J_\perp$, followed by decay of D through the cavity), both $\rho_0(t)$ and $\rho_1(t)$ eventually reach the same steady state. Then $\Delta X(t) \rightarrow 0$ and ΔM saturates.

The phase φ is a free choice; in practice one picks it to maximize $|\Delta M|$.

Does the initial state of D matter? In general, yes. Replace $|g\rangle\langle g|_D$ by an arbitrary ρ_D . How much this changes the result depends on whether D forgets its initial state during the measurement.

- **D relaxes fast.** Given S in $|j\rangle$, suppose D and the cavity relax to a unique steady state (one that still depends on j) on a time short compared with τ_m . Then ρ_D only changes the transient part of $\Delta X(t)$, and the long-time slope Γ_m is independent of ρ_D . This is typical near resonance, where D hybridizes with the cavity and decays quickly through it.
- **D relaxes slowly.** Our model has no intrinsic qubit relaxation. D can only lose its excitation through the cavity (Purcell decay, rate $\approx \kappa g^2/(\omega_D - \omega_r)^2$ in the dispersive regime) or by a swap with S via $J_\perp$. Deep in the dispersive regime this is slow, and σ_z^D is effectively conserved during the measurement. The cavity shift is then proportional to σ_z^D (Sec. 7), so its sign depends on the state of D.

In the second case a mixed detector state, $\rho_D = p_g|g\rangle\langle g| + p_e|e\rangle\langle e|$, leaves the cavity in a *mixture* of coherent states. Two things happen. The signal difference ΔX shrinks, because the two branches of D partly cancel. And the noise grows, because a mixture of coherent states has $C_j \neq 0$: the classical spread between the branches adds to the shot noise in Eq. (16). A thermal or imperfectly reset detector is therefore a source of measurement inefficiency, and a good first test of the excess-noise term. Coherences of D in the σ_z^D basis do not enter $\langle a \rangle$ to leading order in the dispersive regime; near resonance they matter during the transient.

5 The noise

The variance of M is a double integral over the output correlation function:

$$\sigma_j^2(\tau) = \langle M^2 \rangle_j - \langle M \rangle_j^2 = \int_0^\tau \int_0^\tau \left[\langle X_{\text{out}}(t) X_{\text{out}}(t') \rangle_j - \langle X_{\text{out}}(t) \rangle_j \langle X_{\text{out}}(t') \rangle_j \right] dt dt'. \quad (11)$$

The integrand is symmetric under $t \leftrightarrow t'$ (the X_{out} commute), so it is enough to compute it for $t > t'$. Insert $X_{\text{out}} = X_{\text{in}} + \sqrt{\kappa} X_\varphi$ and expand into four pieces.

Step 1: input–input. Only $\langle b_{\text{in}} b_{\text{in}}^\dagger \rangle$ survives:

$$\langle X_{\text{in}}(t) X_{\text{in}}(t') \rangle = \langle b_{\text{in}}(t) b_{\text{in}}^\dagger(t') \rangle = \delta(t - t'). \quad (12)$$

Integrated over the square this gives τ . This is the shot noise: it is there even when nothing is in the cavity.

Step 2: input at the later time t . The term $\langle X_{\text{in}}(t) X_\varphi(t') \rangle$ vanishes. $b_{\text{in}}(t)$ commutes with $X_\varphi(t')$ (causality, $t > t'$) and then annihilates the vacuum on the right; $b_{\text{in}}^\dagger(t)$ annihilates the vacuum on the left.

Step 3: input at the earlier time t' . The term $\langle X_\varphi(t) X_{\text{in}}(t') \rangle$ does not vanish. The $b_{\text{in}}(t')$ part gives zero on the right. For the $b_{\text{in}}^\dagger(t')$ part we use $\langle b_{\text{in}}^\dagger(t') X_\varphi(t) \rangle = 0$ to replace the product by a commutator, and then (5):

$$\langle X_\varphi(t) b_{\text{in}}^\dagger(t') \rangle = \langle [X_\varphi(t), b_{\text{in}}^\dagger(t')] \rangle = -\sqrt{\kappa} \langle [X_\varphi(t), a^\dagger(t')] \rangle. \quad (13)$$

Physically, the noise that entered at t' was partly reflected into the output at t' and partly went into the cavity. It is therefore correlated with the cavity field at the later time t .

Step 4: collect. With the prefactors, for $t > t'$,

$$\langle X_{\text{out}}(t) X_{\text{out}}(t') \rangle = \delta(t - t') + \kappa \left[\langle X_\varphi(t) X_\varphi(t') \rangle - e^{i\varphi} \langle [X_\varphi(t), a^\dagger(t')] \rangle \right]. \quad (14)$$

Write out $X_\varphi = e^{-i\varphi} a + e^{i\varphi} a^\dagger$ in the bracket. The commutator removes exactly the orderings that are not normal-ordered, and the bracket becomes

$$e^{-2i\varphi} \langle a(t) a(t') \rangle + \langle a^\dagger(t) a(t') \rangle + \langle a^\dagger(t') a(t) \rangle + e^{2i\varphi} \langle a^\dagger(t') a^\dagger(t) \rangle \equiv \langle \mathcal{T} : X_\varphi(t) X_\varphi(t') : \rangle. \quad (15)$$

Every $a^\dagger$ stands to the left of every a (normal order, $: :$). Among the a 's the later time stands to the left, and among the $a^\dagger$'s the later time stands to the right (time order, $\mathcal{T}$). This is the Gardiner–Collett result: *output correlations = vacuum δ -function + $\kappa \times$ normally and time-ordered cavity correlations*.

Result. Subtracting the product of means (the δ -term has zero mean) gives

$$\sigma_j^2(\tau) = \tau + \kappa \int_0^\tau \int_0^\tau C_j(t, t') dt dt', \quad C_j(t, t') = \langle \mathcal{T} : X_\varphi(t) X_\varphi(t') : \rangle_j - \langle X_\varphi(t) \rangle_j \langle X_\varphi(t') \rangle_j.$$

(16)

The second term is the excess noise from the system itself. It can be positive (extra fluctuations) or negative (squeezing below shot noise).

Computing C_j : quantum regression. The two-time averages come from the same Lindbladian. For $s \geq 0$,

$$\langle A(t' + s)B(t') \rangle = \text{Tr}\{A e^{\mathcal{L}s}[B\rho(t')]\}, \quad \langle B(t')A(t' + s) \rangle = \text{Tr}\{A e^{\mathcal{L}s}[\rho(t')B]\}. \quad (17)$$

Example: $\langle a^\dagger(t')a(t) \rangle$ with $t = t' + s$ is the second form with $A = a$, $B = a^\dagger$. Propagate $\rho(t')a^\dagger$ for a time s with $\mathcal{L}$, then take the trace with a . In QuTiP this is what `correlation_2op_2t` does.

A useful check. If the cavity is in a coherent state (for each j), all normally ordered moments factorize and $C_j = 0$. Then $\sigma_j^2 = \tau$: pure shot noise.

6 Signal-to-noise ratio, measurement rate and efficiency

With means and variances in hand, the signal-to-noise ratio of the integrated signal is

$$\text{SNR}(\tau) = \frac{|\Delta M(\tau)|}{\sigma_0(\tau) + \sigma_1(\tau)} = \frac{\sqrt{\kappa} \left| \int_0^\tau \Delta X(t) dt \right|}{\sigma_0(\tau) + \sigma_1(\tau)}. \quad (18)$$

Think of two histograms of M , one for each S state, with centres ΔM apart and widths σ_0, σ_1 . $\text{SNR} \approx 1$ means they just start to separate.

Measurement rate. In the QND-like regime, after the ring-up, $\Delta M \propto \tau$ and $\sigma_j^2 \propto \tau$, so SNR^2 grows linearly in τ . The measurement rate is the slope:

$$\Gamma_m = \frac{d\text{SNR}^2}{d\tau} \quad (\text{linear regime}), \quad \tau_m : \text{SNR}(\tau_m) = 1, \quad \tau_m \approx 1/\Gamma_m. \quad (19)$$

The factor convention matters when comparing with papers. With SNR defined as in (18), the quantum limit reads $\Gamma_m \leq 2\Gamma_\varphi$, where Γ_φ is the measurement-induced dephasing rate (Sec. 7).

Detection efficiency. Losses after the cavity (cables, circulators, a non-ideal amplifier) are modelled as a beam splitter of transmission η that mixes in fresh vacuum,

$$X_{\text{det}}(t) = \sqrt{\eta} X_{\text{out}}(t) + \sqrt{1 - \eta} X_{\text{vac}}(t). \quad (20)$$

Redoing the signal and noise with X_{det} : the signal gets a factor $\sqrt{\eta}$, the shot noise stays τ (the two vacua add up to one), and the excess noise gets a factor η . So the prefactor κ in ΔM and in Eq. (16) is replaced by

$$\kappa \rightarrow \eta \kappa, \quad (21)$$

while the κ inside $\mathcal{L}$ is unchanged (the cavity does not know about the losses after it).

Beyond boxcar integration. $M(\tau)$ weights all times equally. A filtered signal $M_w = \int w(t) X_{\text{out}}(t) dt$ obeys the same formulas with $w(t)$ inserted under the integrals: one factor in ΔM , $w(t)w(t')$ in the variance, and $\int w^2 dt$ instead of τ for the shot noise. When the noise is close to shot noise, the best choice is the matched filter $w(t) \propto \Delta X(t)$. This matters mostly in the non-QND regime, where late times carry no information.

7 Worked example: the dispersive limit

In the dispersive limit the recipe gives a closed formula, $\Gamma_m = \eta \kappa |\alpha_1 - \alpha_0|^2$, which reaches the quantum limit $\Gamma_m = 2\Gamma_\varphi$ when $\eta = 1$.

Toy model. Suppose that, after eliminating D, the only effect of S is to shift the cavity frequency, $\Delta_r \rightarrow \Delta_r + \chi_j$ with $\chi_1 = +\chi$, $\chi_0 = -\chi$. The cavity alone then obeys a linear equation.

Where does this come from in our model? Take $J_\perp = 0$ and D far detuned from the cavity, $|g| \ll |\omega_D \pm 2J_L - \omega_r|$. The term $J_L \sigma_z^S \sigma_z^D$ shifts the D frequency to $\omega_D + 2J_L s$, where $s = \pm 1$ is the eigenvalue of σ_z^S . The dispersive D-cavity coupling then shifts the cavity by $\chi_s \sigma_z^D$, with

$$\chi_s = \frac{g^2}{\omega_D + 2J_L s - \omega_r}. \quad (22)$$

With D in $|g\rangle$ (and σ_z^D conserved, see Sec. 4) the cavity shift is $-\chi_s$. The part common to both S states is absorbed into Δ_r , and the toy model's χ is half the difference, $\chi = |\chi_+ - \chi_-|/2$. This is why the toy model needs D in a definite σ_z^D eigenstate: for D in $|e\rangle$ the sign of the shift flips, and for a mixed D the cavity is no longer in a coherent state.

Signal. From the Lindblad equation, $\frac{d}{dt} \langle a \rangle = -[i(\Delta_r + \chi_j) + \kappa/2] \langle a \rangle + \varepsilon$. Starting from vacuum,

$$\langle a(t) \rangle_j = \alpha_j \left(1 - e^{-[i(\Delta_r + \chi_j) + \kappa/2]t} \right), \quad \alpha_j = \frac{\varepsilon}{\kappa/2 + i(\Delta_r + \chi_j)}. \quad (23)$$

After the ring-up ($t \gg 1/\kappa$), $\Delta X = 2\text{Re}[e^{-i\varphi}(\alpha_1 - \alpha_0)]$. The best phase $\varphi = \arg(\alpha_1 - \alpha_0)$ gives $\Delta X = 2|\alpha_1 - \alpha_0|$, and $\Delta M \approx 2\sqrt{\kappa} |\alpha_1 - \alpha_0| \tau$.

Noise. A linear damped cavity driven by a coherent tone stays in a coherent state. So $C_j = 0$ and $\sigma_0^2 = \sigma_1^2 = \tau$.

Rate. Putting this together,

$$\text{SNR}^2 = \frac{4\kappa|\alpha_1 - \alpha_0|^2 \tau^2}{4\tau} \Rightarrow \Gamma_m = \eta \kappa |\alpha_1 - \alpha_0|^2, \quad (24)$$

where we restored η . For a drive at the bare cavity frequency ($\Delta_r = 0$) both states have the same photon number $\bar{n} = |\varepsilon|^2/(\kappa^2/4 + \chi^2)$, and

$$\Gamma_m = \frac{4\eta\kappa\chi^2\bar{n}}{\kappa^2/4 + \chi^2} \approx \frac{16\eta\chi^2\bar{n}}{\kappa} \quad (\chi \ll \kappa). \quad (25)$$

Comparison with dephasing. The measurement-induced dephasing rate for the same toy model is known (Gambetta *et al.*, Phys. Rev. A **74**, 042318 (2006)): $\Gamma_\varphi = \kappa|\alpha_1 - \alpha_0|^2/2$. Therefore

$$\frac{\Gamma_m}{2\Gamma_\varphi} = \eta \leq 1. \quad (26)$$

This ratio is the measurement efficiency. It equals 1 only when no photon is lost on the way to the detector. Photons lost there still dephase the qubit, but they carry information nobody records. (Internal cavity losses, once added, would reduce the efficiency further in the same way.) In this toy model the intrinsic qubit dephasing γ_S adds to Γ_φ but not to Γ_m , so it lowers the efficiency if one defines it with the total dephasing rate; we return to this when we discuss efficiency for the full model.

For our full model the toy model is only the starting point: the cavity sees D, not S, so Γ_φ must be computed for S itself. That is the subject of the next set of notes.

8 Beyond the dispersive limit, and the recipe

Nothing in Secs. 4–6 used the dispersive approximation. The same formulas hold near resonance and with the flip-flop coupling. Only two things change.

- **Excess noise.** When the cavity hybridizes with D (resonant regime, or strong drive), the cavity is no longer in a coherent state. Then $C_j \neq 0$ and must be computed with quantum regression. It can raise or lower the noise relative to shot noise.
- **Saturating signal.** With $J_\perp \neq 0$, S relaxes (through D and the cavity) during the measurement and $\Delta X(t)$ decays. $\text{SNR}^2(\tau)$ then has a maximum instead of growing linearly. The useful numbers become the maximum SNR, the time at which it is reached, and the time τ_m at which $\text{SNR} = 1$ (if it is reached at all).

One caveat: the SNR assumes roughly Gaussian histograms. Near resonance they may not be. The SNR is then still a well-defined figure of merit for a linear detector, but not necessarily the best one. Going further requires the stochastic master equation, which we postpone.

Recipe.

1. Choose parameters and build $\mathcal{L}$ for S + D + cavity (truncate the cavity Fock space well above the largest photon number).
2. For $j = 0, 1$, solve the Lindblad equation from $\rho_j(0)$ and store $\langle a(t) \rangle_j$.
3. Form $\Delta X(t)$ and pick φ to maximize $|\Delta M(\tau)|$.
4. For each j , compute the two-time correlations in $C_j(t, t')$ by quantum regression, and the double integral (16) for $\sigma_j^2(\tau)$.
5. Assemble $\text{SNR}(\tau)$, with the prefactor $\kappa \rightarrow \eta\kappa$. Read off Γ_m from the slope of SNR^2 (QND-like), or τ_m and the maximum SNR (non-QND).
6. Check: in the dispersive, far-detuned limit with $J_\perp = 0$, you should recover $\Gamma_m = \eta\kappa|\alpha_1 - \alpha_0|^2$ and $C_j \approx 0$.